Goal-Conditioned Activity-Profile Synthesis: Agent Search, Random Baselines, and Windowed Gate Validation

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Abstract

Can an agent construct legal hardware workloads that approach a requested activity profile, rather than maximize stress? We implement a typed, proposal-budgeted harness and evaluate a frozen Gemini controller against classical search on AES and AXI DMA. Two development/held-out protocols cover 710 cells and 32,620 charged proposal slots. The scalar agent has a small, nonsignificant numerical advantage over random on AES and a disadvantage on DMA. In a fresh structural/temporal study, random has the lowest mean AUC on both designs; the AES agent is significantly worse after multiplicity correction. These results do not establish agent superiority or equivalence. We separately validate sixteen predeclared temporal finalists using matched functional gate simulation and native eight-window OpenSTA activity reads. This separates power shape from nearly identical full-window means. The contribution is a reproducible evaluation and measurement method, with a bounded negative finding about the tested controllers—not a claim that agents cannot synthesize useful hardware workloads.

1. Question and scope

Hardware activity depends on stimulus, but finding a workload for an arbitrary requested behavior differs from finding a maximum-power stressmark. Automated stressmark generation predates LLMs [1]; surrogate-assisted genetic search also targets CPU power viruses [2]. We therefore make no first-to-generate directed-power-workloads claim. Our question concerns target-conditioned search under a common legal representation and an auditable budget.

The original motivation was that semantic reasoning might help with long, design-specific action sequences. The experiments here test a narrower question: can a controller choose effective bounded scalar or structural edits when an adapter supplies the low-level implementation? All methods receive the same workload language. Expressiveness belongs to that language; an agent does not gain expressiveness merely by using it.

The measured search endpoint is RTL transition activity, not watts. Gate power is a post-search validation tier. We retain this distinction even when a selected workload's gate profile follows its reference. Such evidence does not retrospectively make the search power-guided or validate a universal activity-to-power predictor.

2. Interface and evaluation

Each design adapter defines a typed workload schema, protocol constraints, deterministic lowering, functional checks, and useful work. Schema and protocol checks precede simulation; functional correctness and a hard useful-work gate follow it. Invalid workloads receive no power score. Rejected, missing, and duplicate proposals still consume budget. This prevents cheap malformed outputs from obtaining unlimited retries. Simulations, tokens, cost, tool identities, and source hashes are recorded separately.

The scalar adapter exposes workload parameters. The structural adapter adds bounded sequence/repeat/pacing schedules and shared structural edits. AES executes 64 blocks over 6,774 clock edges; DMA completes 4,096 bytes over 12,000 edges. Each structural contract includes 6,000 idle cycles. Work and observation horizon are fixed, so temporal shape cannot be improved simply by shortening the measurement or eliminating useful computation. DMA uses fixed-size copies with pacing/concurrency, not its entire transfer-length and backpressure space.

The model receives a common template containing design metadata, the schema, constraints, goal, and compact feedback. In the temporal study, four shared random proposals initialize each run. The agent then gets seven feedback rounds of four proposals; the hybrid alternates four model rounds with three CPU rounds. This is closed-loop search, not a one-shot test. The frozen configuration uses Gemini 2.5 Flash, temperature 0.7, top-p 0.95, 512 thinking tokens, and an 8,192-token output limit. A malformed batch is charged, not silently resampled until it succeeds.

Reusable simulator, activity, synthesis, and power components are separated from research drivers, following CHIA's composable-loop motivation [3]. The reported studies ran through host Python drivers; they are not evidence of a deployed Ray/CHIA execution or a successfully replayed container artifact.

Frozen comparisons: primary endpoints first

3. Development and held-out protocols

The scalar protocol selects one controller on a balanced development panel, then freezes it before seeds 200–209. AES has six policies and DMA five; each uses five targets $q \in \{.10, .25, .50, .75, .90\}$, 50 proposal slots, batch size four, and tolerance .02. The resulting 550 cells contain 27,500 slots. Calibration and implementation hashes are archived; historical pilot results with different envelopes are not pooled with them.

The later structural study uses fresh seeds 400–409, two achieved reference profiles per design, four policies, and 32 slots per run: 160 cells and 5,120 slots. Development selected the best-eight population family. A diversity-preserving population ablation did not win selection. This was inspired by population-based agent search, not a reproduction of AlphaEvolve or evolution of arbitrary generator programs. Targets were observed in development: fresh seeds are not unseen-target generalization.

The primary endpoint is trapezoidal area under the best-so-far error curve over charged proposal indices. Scalar error is envelope-normalized; temporal error is the predeclared capped, fixed-scale NRMSE (scales 200 for AES and 40 for DMA). Temporal tolerance is .10. Unsolved runs remain right-censored, contributing 50 or 32 to capped evaluations-to-target. That cap is not an estimate of eventual solution time.

For inference, target-paired AUC differences are averaged within each seed, leaving ten seed units. We use 10,000-replicate paired bootstrap intervals and exact two-sided sign-flip tests. Holm correction covers nine scalar or eight temporal comparisons, separately. Intervals are pointwise, not simultaneous. With ten units, these tests have coarse resolution; failure to reject is not equivalence.

4. Scalar results

Policy	AES AUC	DMA AUC
Random	2.2393	2.0849
Mutation	4.5159	4.0012
Evolutionary	4.5674	3.2367
Scalar-edit	3.5316	6.5367
Coverage-guided	3.3627	—
Agent	2.1938	2.6507

The AES agent–random difference is -0.0455, CI $[-0.6579, 0.6310]$, Holm $p = .8965$. DMA's difference is $+0.5658$, CI $[0.0787, 1.0669]$, Holm $p = .1992$. Only DMA comparisons against mutation and scalar-edit support agent superiority. The latter baseline is valid in only 22.92% of slots, which limits the importance of that win. Random is the stronger comparator. Agent/random solve counts are 43/38 out of 50 on AES and 36/38 on DMA.

5. Structural/temporal results

Design	Policy	AUC	Solved
AES	Random	3.3125	15/20
	Evolutionary	3.7610	12/20
	Agent	3.7027	8/20
	Hybrid	3.6745	9/20
DMA	Random	2.4924	20/20
	Evolutionary	2.6550	18/20
	Agent	2.7371	14/20
	Hybrid	2.7784	18/20

Random has the lowest mean AUC on both designs. The AES agent–random difference is $+0.3902$, CI $[0.2107, 0.5703]$, Holm $p = .03125$: the baseline is better. None of the other seven corrected comparisons supports superiority in either direction. In particular, hybrid search does not establish an advantage over random or evolution. These are results for the specified controller, grammar, targets, and budgets, not random-search optimality.

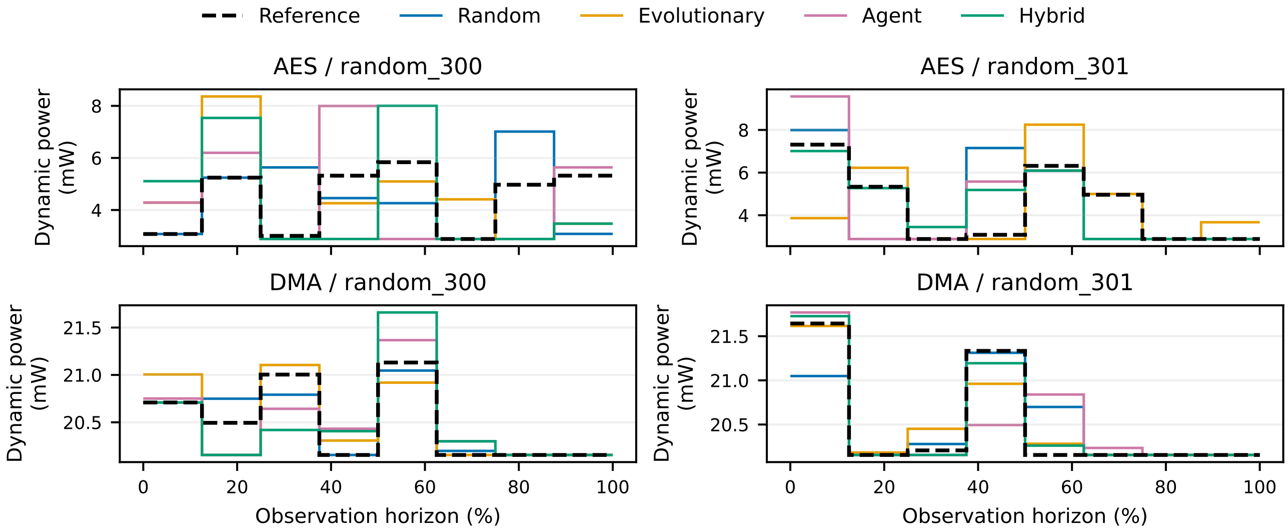
6. Compliance, feedback, and cost

Temporal agent validity is 562/640 on AES and 573/640 on DMA; random and evolutionary are 640/640. All temporal failures are labeled SCHEMA, but this includes unknown edit operations, illegal field sets, type mismatches, redistribution errors, and missing outputs. It does not mean every failure was malformed JSON. Zero downstream failures cannot establish that the model understood all hardware constraints: invalid edits were rejected upstream, and the adapters supplied much of the legality machinery. The scalar study also contains protocol and useful-work failures.

A post-hoc equal-valid-evaluations analysis retains only paired cells that reach a chosen valid budget within the original proposal budget. At $K = 16$ all 20 temporal pairs per design survive: agent/random mean curve errors are .12811/.11946 on AES and .10168/.09593 on DMA. The descriptive ordering does not reverse. The hybrid also remains above random. This diagnostic does not replace primary AUC or isolate a causal compliance effect: failures already changed adaptive history, and larger K can select a nonrepresentative subset.

Recorded configured-rate model cost is approximately \$4.3444 for scalar and \$1.5012 for temporal, with eight and two unknown-usage batches respectively. These are incomplete usage estimates, not reconciled cloud bills. Temporal usage records 2,431,745 input and 308,664 output tokens. CPU policies incur no LLM charges, not zero compute cost. Summed temporal evaluation and proposal-generation times are 7,591 and 2,665 seconds; shared-host concurrency precludes interpreting these as controlled speed comparisons. No larger-model study was added after inspecting the held-out outcomes.

From full-window averages to temporal gate profiles



7. Matched gate measurement

The best valid seed-400 trial per design/target/policy was selected without gate scores; ties use earliest proposal. All sixteen passed matched functional GLS. Unmasked AES has 90,247/90,247 annotated pins; DMA has 36,292/36,296 (four missing sideband pins). Configuration, stimulus order, work, and each waveform's own horizon are reconciled. Zero-delay models omit timing glitches.

Historical RTL-to-netlist annotation recovered only 203/154,059 pins in a different AES configuration; this is not a matched comparison with current unmasked AES. Legacy correlations contaminated by configuration or window normalization are superseded, not supporting evidence.

Full-window means are nearly identical: 4.4572 mW on AES and 20.495 mW on DMA. Equal means cannot distinguish shape, although fixed work and duration alone do not mathematically guarantee equal energy.

We use pinned upstream OpenSTA [4] native VCD time bounds and retain the original waveforms. Eight bins follow the frozen rising-edge sample partition. Internal boundaries fall one VCD tick before the indexed edge; we reject any boundary containing a waveform timestamp. Inclusive native endpoints therefore cannot double-count a transition. Each waveform's own timescale and exact duration are retained, including reset. RTL and GLS grids must match, and all bin durations must sum to the original span.

8. Independent checks and endpoint

Known traces at 1 ps and 10 ps timescales independently check transition counts, carried-in state, high-time fractions, and boundary ownership. Fresh OpenSTA processes prevent stale annotation between windows. Every read records internal, switching, leakage, total power, and annotation. Dynamic power is internal plus switching, not total including leakage.

The first case exposed float32 design-total accumulation: weighted switching differed from full-window switching by 20.78 ppm. Independently summing the same 22,901 leaf powers in double precision reduced the discrepancy to 0.016 ppm. The unchanged 10^{-5} relative additive check uses the leaf sum; reported profiles retain the native design totals. Conditional internal-power estimation need not be additive in averaged activity, so its discrepancy is reported rather than forced away.

The four achieved schedules are replayed through GLS to define watt-valued targets R_i. For finalist P_i and duration d_i, the descriptive metric is

$$E_{\text{gate}} = \sqrt{\frac{\sum_i d_i (P_i - R_i)^2}{\sum_i d_i R_i^2}}.$$

The denominator is reference-only; candidate-peak normalization is not used. This gate error is distinct from the frozen activity loss. There is no new gate solved threshold or policy significance test on selected finalists.

Selected-case findings

All 20 waveforms pass the window checks. AES finalist bins range from 2.887 to 9.569 mW; DMA bins range from 20.155 to 21.769 mW. Gate NRMSE spans .225–.404 on AES and .00346–.01887 on DMA. Within each target/design group, finalist activity-error ordering is preserved in gate error. This is descriptive selected-case agreement, not universal proxy validity. DMA's large common dynamic baseline reduces relative error; small NRMSE is not proof of exact shape matching. The figure shows all sixteen candidates and four references, with each tier's matched measurement horizon.

Interpretation, limitations, and reproducible evidence

9. What the experiments establish

The scalar and structural protocols independently fail to establish a general agent advantage over random, with a corrected significant agent loss on temporal AES. This narrows the empirical hypothesis; it does not show that agents cannot create useful schedules. They generated valid finalists and often solved activity targets. The missing result is superiority over equally expressive, properly budgeted baselines.

Strong random search and weaker local/evolutionary refinement are consistent with broad sampling being effective in these bounded spaces. They do not prove that activity is monotonic in block count, that random is near-optimal, or that hardware size explains the ordering. The temporal experiment fixes work and duration, so the scalar block-count explanation cannot simply be carried over to it. A causal account requires additional controlled ablations.

The legal adapter is both a strength and a boundary. It makes comparisons reproducible, enforces hard anti-idle constraints, and reduces implementation errors. It may also remove some long protocol-prerequisite reasoning that motivated the original hypothesis. That possibility is an interpretation, not a measured finding that fairness "designed the hypothesis away." A shared grammar does not forbid semantic advantage; it lets an advantage in search, if present, be distinguished from privileged access to a better API.

10. Limits and future work

Only two designs enter the frozen comparative studies. Ibex infrastructure and pilots are not substituted for a completed third-design panel. Compositional goals are not evaluated here. The temporal targets are achieved development profiles, not arbitrary unseen requests. Thirty-two or fifty slots test bounded search, not unlimited reasoning; Flash is one model configuration. These limits preclude a blanket conclusion about all agents, larger models, deeper search, or complex SoCs.

More demanding protocol-depth tasks, generator-program evolution, or larger-model/deeper-budget studies remain plausible future experiments. They require fresh development and held-out evaluation, not retuning this observed panel until an agent wins. Classical baselines must receive the same representation and evaluator budget. A development-only probe would be exploratory evidence, not an extension of the confirmatory result.

Windowed validation is descriptive evidence on selected cases. It cannot establish universal RTL-to-gate prediction, power-directed optimization, silicon accuracy, or signoff power. Sky130 Liberty characterization and complete annotation do not supply parasitics or timing-induced glitches. Near-identical full-window means do not imply identical temporal profiles; conversely, a low selected-case gate error is not an independent policy win.

11. Artifact and reproducibility

The repository archives compact ledgers, source/configuration manifests, frozen calibration and target corpora, paired inference, selected workload identities, raw power reports, and hashes. Large VCDs remain scratch artifacts; replay commands and their hashes are provided rather than committing waves. The four-page PDF and its source are included with data-derived figures.

The scalar, temporal, and windowed reports under `docs/` link their authoritative archives. Window measurements are versioned separately, so no later oracle change rewrites a frozen search result. Local audits verify ledgers and regenerate descriptive reports without model calls.

The current evidence comes from host execution. A container configuration exists, but its latest validation smoke failed before Python started; a successful full image replay is not claimed. Completing that portability check, venue formatting, and the upstream contribution remain submission work, not hidden completed deliverables.

Conclusion

Within two frozen, bounded workload studies, a feedback-driven agent does not establish general superiority over strong random search. The artifact makes that result inspectable and separates it from gate-measurement validity. Matched native-window validation addresses temporal power shape without confusing activity, total energy, and full-window mean power. The appropriate next scientific question is where semantic search helps beyond these bounded adapters, not how to relabel the observed losses as wins.

References

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[3] A. Cui et al. *CHIA: An open-source framework for principled, agentic AI-driven hardware/software co-design research.* 2026. arXiv:2606.27350v3.

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